# Meteorological Ensemble Postprocessing and Site-Level Extreme Weather Decision Kernels

A research demonstrator proposal for the paceval. Mathematical Engine

paceval. Industrial Research Portfolio

14 November 2026

**Abstract**

This whitepaper specifies a reproducible research demonstrator that transforms external numerical weather prediction ensembles into site-level probabilities, loss estimates and action margins for industrial infrastructure. The proposed contribution is not a new atmospheric forecast model. It is the repeated execution and independent verification of explicit mathematical calibration and decision functions using the paceval. Mathematical Engine. Rainfall, wind-gust and frost hazards are treated as distinct events with precisely defined observation windows. The design uses publicly documented forecast and station-data sources and is benchmarked against optimized native numerical evaluation. No forecast-skill gains, runtime speedups or operational warning capabilities are claimed in advance.

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## Executive summary

**Research question.** Can reusable paceval mathematical execution graphs evaluate large volumes of calibrated, site-specific ensemble weather risk functions accurately, efficiently and transparently, while preserving the scientific integrity of the original forecasts?

**Proposed use case.** A critical-infrastructure operator monitors thousands of assets. On every forecast update, the operator receives ensemble weather model outputs and must calculate the probability of a defined local event, estimate whether an optional protective action has positive expected value, and explain how the mathematical result was obtained. The primary prototype addresses **24-hour heavy precipitation, interval-specific extreme wind gusts, and minimum 2-metre air temperature below freezing**. These are forecast-defined hazards. They are **not direct predictions of flooding, structural failure or surface icing**.

**Why paceval.** Most numerical weather prediction effort belongs in specialised atmospheric models and HPC systems. The target for paceval is the repeated calculation of explicit, parameterised statistical and decision expressions once data have been ingested and standardised. The paceval API supports reusable computation handles, batch result evaluation and optional Trusted Interval Computation (TINC) for the arithmetic evaluation layer [R17, R18]. The proposed evaluation asks whether that architecture improves maintainability, auditability and potentially mathematical-kernel throughput under fair comparisons.

**Scientific position.** Forecast probability calibration is already an established meteorological field. Ensemble Model Output Statistics (EMOS), forecast verification and local postprocessing are not claimed as inventions of this research [R11, R12, R13]. The distinctive demonstration is a reusable, versioned paceval execution layer with explicit mathematical contracts, batch processing, independent references and decision-cost sensitivity.

**Evidence status, as of 14 November 2026.** The models, experiments and acceptance criteria in this paper are a **research plan**, not completed numerical measurements. All numerical business-loss values shown in the decision illustration are synthetic examples.

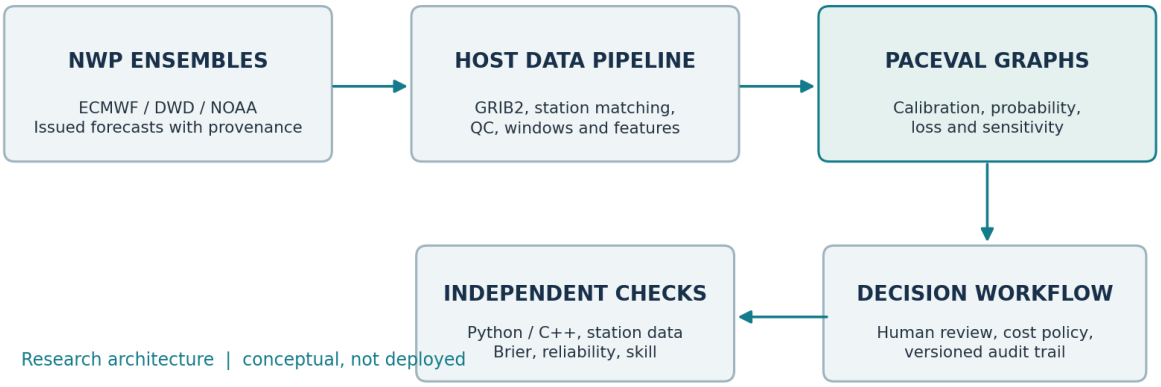



Figure 1: Proposed data and execution architecture. This is a conceptual diagram, not a deployed system.

# 1. Industrial problem and value proposition

Weather-dependent operations often need decisions for individual facilities rather than regional forecast maps. Examples include preparing industrial sites for heavy rainfall, securing outdoor equipment before extreme gusts, and adjusting a planned outdoor operation during a predicted frost window. An operator must evaluate many location, lead-time, model-cycle and asset combinations, then explain the threshold and cost assumptions to technical staff.

Existing numerical weather prediction centres already provide ensemble output and probabilistic products. ECMWF's IFS ensemble uses 50 perturbed members and one control member [R2]. ECMWF open datasets include IFS and AIFS products in GRIB2, with a subset now also available on a finer 0.1-degree IFS output grid from October 2026 [R1]. NOAA GEFS and the DWD ICON ensemble product families offer alternative data pathways [R4, R5]. The NWS explicitly frames weather guidance around decision impacts and probabilistic risk communication [R10].

**Target customer profiles** are industrial asset operators, utilities, transport and logistics planners, industrial-site emergency planners and specialist meteorological service companies. This document makes no forecast-skill, cost-saving or market-size claim about any individual customer.

## 1.1 Why this is distinct from existing paceval projects

The earlier industrial portfolio already includes wind-farm loss functions, digital twins, industrial gateway analytics, numerical risk scoring and matrix sensor fusion. The new contribution is **meteorological ensemble data with valid-time and accumulation semantics, calibrated rare-event probabilities, spatiotemporally independent forecast verification, and action-value calculations**. It therefore remains distinct from direct wind-energy output estimation or generic anomaly scoring.

## 1.2 Demo scope and intentionally excluded scope

| Included in the proposed demonstrator | Outside the first demonstrator |
|---|---|
| Ingest prepared historical or live ensemble features | Running ECMWF IFS, ICON or GEFS atmospheric simulation |
| Evaluate raw member exceedance, calibrated probability and sensitivity | Numerical weather prediction, data assimilation and radar nowcasting |
| Compute site-specific action margins and expected loss | Predicting physical flood depth, slope failure or asset failure without further models |
| Process many locations, event windows and asset policies | Issuing authoritative public weather alerts or replacing forecasters |
| Verify probability skill against independent station observations | Claiming statutory warning, safety certification or zero false alarms |

# 2. Weather data and observational evidence

The experimental pipeline should support at least two pathways. **Path A** is a long-term retrospective experiment using a genuinely archived forecast ensemble and corresponding station observations. **Path B** is a Germany-centred live replay using DWD ICON ensemble output and DWD station observations. The second path requires accumulating forecasts over time and cannot substitute immediately for a multi-year hindcast.

| Source | Intended role | Essential qualification |
|---|---|---|
| ECMWF open IFS / AIFS data [R1, R3] | Contemporary ensemble and optional AI-forecast comparison | Open-data access is a rolling archive of recent runs, not a complete historical training archive. Product coverage and grids must be checked for each parameter. |
| ECMWF ensemble documentation [R2] | Ensemble-member conventions and medium-range context | Model-cycle coverage and available lead times differ across streams. |
| NOAA GEFS [R4] | Historical ensemble forecasts for retrospective experiments | Coverage, version, member count and forecast-field availability must be checked for every study period. |
| DWD ICON ensemble directory [R5] | Regional near-real-time ensemble forecasts | Native grid, forecast cycles, variable availability, archive retention and licence must be documented. |
| DWD Climate Data Center [R6] | Germany station observations | Quality-control status, station relocations, missing values and time resolution matter. |
| NOAA Global Hourly [R7] | Historical station observations | Station coverage and variable reporting frequency vary. Missing maxima or accumulations cannot be reconstructed by assumption. |
| Copernicus ERA5 [R8] | Context, climatology or secondary sensitivity analysis | Reanalysis is not a historical issued-forecast archive. It must not be mislabelled as an operational forecast hindcast. |

**Concrete variables.** ECMWF's documented parameters include `tp` (total precipitation), `10fg` (10-metre wind gust), and `2t` (2-metre temperature), with separate probability products at specific thresholds [R3, R19]. Weather measurements require explicit unit conversion: temperature from kelvin to degrees Celsius; precipitation to millimetres from the source's unit; gusts in metres per second. It is not safe to assume that every product reports the same accumulation interval, maximum interval or valid-time convention.

**Operational warning.** The ECMWF open-data portal retains only the most recent forecast cycles for a short rolling period. A statistically meaningful, out-of-time calibration evaluation requires a legitimately acquired historical forecast archive, not merely ERA5 observations or a short scrape of the live portal [R1, R8].

## 2.1 Required input contract

For each site $s$, issue time $t$, lead time $\ell$, hazard $h$ and ensemble member $m$, record:

- Model name, model version, cycle UTC, forecast valid time, ensemble member ID and forecast step-range metadata.
- Site ID, coordinates, elevation, interpolation method, grid location and documented representativeness limitations.
- Forecast variable, original units, converted SI units, averaging or maximum window, and precipitation accumulation semantics.
- Available-member count $N$ after field-level quality checks. Never silently replace a missing member with zero.

- Observation station, observation QC flag, event verification interval and any temporal or spatial matching restriction.
- Immutable mathematical model version, calibration dataset fingerprint and cost-policy version.

Precipitation accumulation differencing, spatial interpolation, data provenance, station matching and missing-data handling belong in the host pipeline. They are not evidence of paceval's mathematical throughput.

## 3. Hazard definitions and event windows

The illustrative demonstration defines three binary events, each with its **own valid-time window**. Threshold values are research defaults only and must be confirmed for a specific operator, geography and asset.

| Hazard | Example event $Y_h = 1$ | Forecast input | Main risk of misinterpretation |
|---|---|---|---|
| Heavy precipitation | $P_{24h} \geq 20\,\text{mm}$ within a precisely defined 24-hour interval | Ensemble precipitation total over that exact interval | High rainfall does not automatically mean flooding. Drainage and catchment state are not modelled. |
| Extreme wind gust | $G_{6h} \geq 25\,\text{m}\,\text{s}^{-1}$ maximum over a defined 6-hour interval | Compatible ensemble maximum gust for the same interval | A 10-metre forecast gust is not a local structural load calculation. |
| Frost-related air-temperature event | $T_{\text{min},overnight} < 0\,^{\circ}\text{C}$ within a defined overnight interval | Compatible 2-metre minimum-temperature output or assembled time samples | Air temperature below zero is not equivalent to road-surface or equipment icing. |

For a binary hazard $h$, the forecast label $Y_{s,t,\ell,h}$ must be derived from **observations of the event itself**, not from a model-analysis field that may share assumptions or assimilation inputs with the forecast. If a station lacks the relevant maximum or accumulation, the case should be marked unavailable, not treated as a negative event.

## 4. Mathematical model and reusable execution kernels

### 4.1 Raw ensemble exceedance and robust input features

For a defined event and valid window, let $k_h$ be the number of ensemble members satisfying the event criterion, out of $N$ valid members:

$$q_h^{\text{raw}} = \frac{k_h}{N}, \qquad q_h^{\text{smooth}} = \frac{k_h + \frac{1}{2}}{N+1}, \qquad u_h = \log\left(\frac{q_h^{\text{smooth}}}{1 - q_h^{\text{smooth}}}\right).$$

The half-count smoothing provides a finite logit when no members, or all members, exceed the threshold. It is a **numerical regularisation**, not proof of calibration or independent Bernoulli trials. Ensemble members are generally dependent and should not be interpreted as $N$ independent observations of nature [R9, R12].

The host computes threshold counts and basic ensemble statistics with validated loops. A subsequent research variant may construct larger paceval expressions for means, variances, normalisations and candidate-score composition, where that creates meaningful graph reuse. The

primary benchmark must distinguish the costs of host-side counting and feature preparation from the costs inside paceval.

## 4.2 Site-specific probabilistic calibration

The initial research model is a constrained, interpretable logistic calibration. For each event class and lead-time group, let $x$ be a validated feature vector:

$$z_h = \beta_{0,h} + \beta_{1,h}\nu_h + \beta_{2,h}\frac{\bar{x}_h - \theta_h}{s_h} + \beta_{3,h}\frac{\sigma_h}{s_h} + \beta_{4,h}\frac{\Delta z_s}{s_z} + \beta_{5,h}\sin\left(\frac{2\pi d}{365.25}\right) + \beta_{6,h}\cos\left(\frac{2\pi d}{365.25}\right),$$

$$p_h = \frac{1}{1+\exp(-z_h)}.$$

Here $\bar{x}_h$ and $\sigma_h$ are relevant ensemble mean and standard deviation; $\theta_h$ is the event threshold; $s_h > 0$ and $s_z > 0$ are training-set scale factors; $\Delta z_s$ is the site-to-grid elevation difference; $d$ is the day of year; and $\beta$ coefficients are fitted exclusively on past data. Both units and feature definitions must be versioned. For precipitation, a robust alternative uses a log-transformed positive accumulation rather than a naive Gaussian assumption. Separate hazard-specific features should be chosen before fitting.

This is a **candidate model**, not a claim that one logistic form is optimal for all hazards. Compare it with raw ensemble exceedance and suitable established postprocessing methods, including EMOS or other monotonic calibrators when compatible with the target event [R11, R12, R13]. Predictors, lead-time groups and regularisation must be selected inside the training folds. Evaluate model stability on new seasons and held-out regions.

A numerically stable sigmoid formulation or a verified bounded-logit contract is required. The prototype must detect invalid, nonfinite or out-of-domain inputs. It must not silently clip probabilities to disguise arithmetic errors.

## 4.3 Evaluation losses, sensitivity and attribution

For an observed binary outcome $y_i \in \{0, 1\}$, the per-case Brier loss is:

$$B_i = (p_i - y_i)^2, \qquad \mathrm{BS} = \frac{1}{n}\sum_{i=1}^{n} B_i, \qquad \mathrm{BSS} = 1 - \frac{\mathrm{BS}_{\mathrm{model}}}{\mathrm{BS}_{\mathrm{reference}}}.$$

BSS is undefined if the reference Brier score is zero. Raw-model and climatological reference scores should be reported separately, not silently substituted. An interpretable local derivative, holding other inputs fixed, is:

$$\frac{\partial p_h}{\partial \nu_h} = \beta_{1,h}\, p_h(1 - p_h).$$

The calibrated probability, Brier contribution, decision margin and selected analytic sensitivities are suitable reusable paceval expressions. Aggregation, reliability-bin formation and confidence-interval estimation remain orchestration tasks unless a separately benchmarked paceval graph is justified.

**Calibration quality is not equivalent to probability sharpness or event discrimination.** Verification must report reliability diagrams, event frequency, uncertainty intervals and rare-event diagnostics in addition to Brier skill [R9, R14].

### 4.4 Expected-loss and action-value kernel

Suppose a protective action costs $C_A \geq 0$ and reduces the otherwise expected loss $L_h \geq 0$ from one event by an estimated fraction $e_h \in [0, 1]$. The action's expected net benefit is:

$$M_h = p_h e_h L_h - C_A.$$

A model-driven suggestion is economically favourable under these assumptions only if $M_h > 0$. If $e_h L_h > 0$, the equivalent probability threshold is $p_h > C_A/(e_h L_h)$. A zero or negative denominator, policy prohibition or unknown mitigation efficacy requires a separately defined policy outcome, not blind evaluation.

**Example (synthetic, not measured):** If an action costs 1,200 units, a weather event causes 10,000 units of avoidable loss and the action prevents 70% of that loss, the model's action threshold is $p_h > 0.1714$. This illustrates why the optimal action probability depends on asset-specific consequences, not only a generic meteorological threshold.

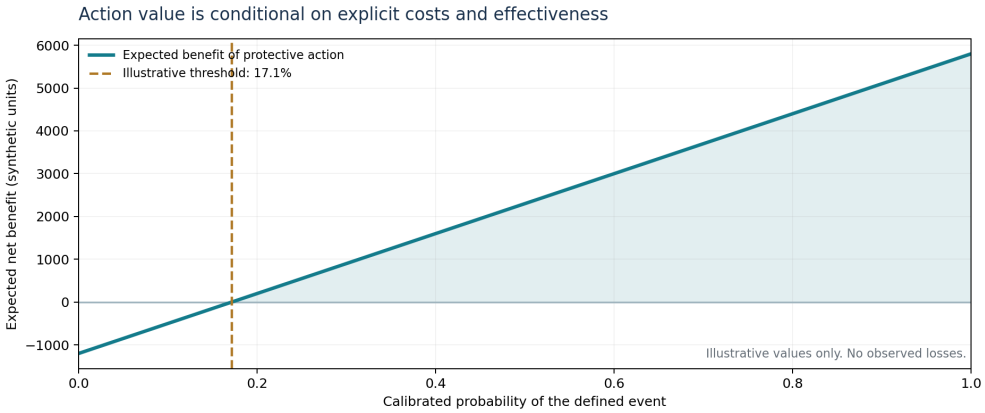



Figure 2: Illustrative expected net benefit versus calibrated hazard probability. The values are synthetic and do not represent observed infrastructure losses.

For additive event losses, expected loss across hazard categories can be summed by linearity of expectation without assuming independence. **Compound-event probability cannot generally be obtained by multiplying marginal probabilities.** If the joint probability is required, calculate joint member exceedance on matched members and matched time windows, then separately assess calibration and dependence. Otherwise report the Fréchet bounds:

$$\max(0, p_A + p_B - 1) \leq P(A \cap B) \leq \min(p_A, p_B).$$

ECMWF's forecast guidance explicitly warns against combining probabilities for different time windows or events without respecting dependence [R15].

## 5. Precise division of responsibilities

| Processing stage | Recommended implementation | Reason |
|---|---|---|
| GRIB2 decoding, station ingestion, UTC alignment | Host C++ and established meteorological readers | I/O, metadata, temporal and geographic correctness are not mathematical-expression kernels. |
| Ensemble member pairing, threshold counting, QC | Host C++ | Requires categorical logic, field checks and stateful alignment. |

| Processing stage | Recommended implementation | Reason |
|---|---|---|
| Normalisation, stable calibration formula and derived probabilities | Reusable paceval graphs | Explicit mathematical expressions with reused coefficients and many input sets. |
| Decision margins, Brier terms, regularisation terms and analytic sensitivities | Reusable paceval graphs, including combined expressions | Makes mathematical use of paceval substantive, versioned and independently testable. |
| Model fitting and parameter search | Host optimizer, optionally evaluating candidate objective graphs through paceval | Iterative workflow and data splits remain host-owned; high-volume candidate mathematics can be delegated. |
| Independent numerical reference | Separate native C++ and Python evaluation | Must not repeat the same paceval implementation as its own oracle. |
| Alerts, dashboards, action approval, provenance | Host application | Business policy, safety permissions and communication need separate control. |

## 5.1 Exact SDK interaction to investigate

The documented paceval API supports `paceval_CreateComputation` for creating reusable computation handles, `paceval_dGetComputationResult` for one numeric input set and `paceval_dGetComputationResultExt` for multiple input sets [R17]. The research demonstrator should use the genuine SDK and link the real paceval binary. A nominal paceval wrapper around independently computed C++ probabilities would not satisfy the experiment.

A minimal mathematically meaningful example, using six validated numerical inputs and an already trained model, is the probability expression:

```
1/(1+exp(-(b0+b1*u+b2*x+b3*spread+b4*height+b5*season)))
```

For a fairer, more demanding demonstration, create a **combined versioned graph** that returns action margin directly from the predictors and asset parameters:

```
(eff*loss)/(1+exp(-(b0+b1*u+b2*x+b3*spread+b4*height+b5*season)))-cost
```

The exact parser notation and implemented function availability must be checked against the SDK version. The formulas above use standard arithmetic and `exp`, without relying on hypothetical built-in weather statistics or undocumented vector functions. Coefficients, features and business parameters should be provided as explicit input variables so graph creation is amortised over many evaluations. Where multiple output values are needed, use multiple persistent graph handles or the documented multiple-computation APIs rather than pretending one scalar function returns a vector.

A representative execution sequence is:

```
1. Validate and version the formula and input contract.
2. Create persistent paceval computation handles once per formula version.
3. Decode and align ensemble and observed weather data in the host.
4. Build validated numeric feature arrays for many sites and forecast cycles.
5. Evaluate persistent handles in batches through the paceval SDK.
6. Check each result and error status. Compare independent reference outputs.
7. Store probability, action margin, model version and audit metadata.
8. Delete handles when no longer needed.
```

### 5.2 TINC interpretation

TINC may be used to investigate numerical intervals around a **specific arithmetic evaluation** as supported by the SDK [R18]. It must not be described as a confidence interval for the meteorological forecast or as a proof of correctness of the fitted calibration. Parameter uncertainty, station representativeness, rare-event sampling error and future forecast errors require separate statistical treatment. Test TINC and independent high-precision references on boundary cases, and report whether the tested SDK actually encloses the intended arithmetic result under stated assumptions.

## 6. Experimental design and realistic benchmarks

### 6.1 Test cohorts and comparisons

1. **Synthetic mathematical smoke test.** Fixed input vectors for $q$, $\nu$, $z$, $p$, $M$ and Brier contribution. Include $k = 0$, $k = N$, near-zero spread, threshold ties, missing-member rejection, extreme valid logits and nonfinite data.
2. **Retrospective forecast test.** Multiple years of archived forecast issues and matching observations where possible. Use genuinely out-of-time test seasons, plus separate held-out stations or geographical regions.
3. **Live operational replay.** Run repeated forecast cycles with documented source availability, missing product handling, duplicate-cycle suppression and batch sizes.
4. **Compute-only benchmark.** Compare the same formulas using paceval persistent graphs, paceval batch evaluation, optimized native C++ scalar evaluation and an appropriately optimized vectorized reference. Use exactly identical numerical inputs, precision, output definitions and error tolerances.
5. **End-to-end benchmark.** Include data decode, spatial interpolation, feature preparation, graph creation if required, evaluation, aggregation, audit logging and output serialization. Report how much total time is actually spent in the paceval stage.

Example load for reproducibility planning only: **10,000 sites, four valid-time windows and three hazard classes**, corresponding to 120,000 site-window-hazard evaluations per cycle. Ensemble-member processing is separately measured and depends on available member counts and the number of model cycles. The number is a proposed workload, not a measured performance result.

### 6.2 Weather forecast skill and reliability

For each hazard, report the following on held-out forecast issues:

| Metric | Why it matters | Scientific guardrail |
|---|---|---|
| Brier score and Brier skill score | Overall mean squared probability error versus chosen baseline | A low Brier score can conceal near-zero event frequency. |
| Reliability diagram with bin counts | Consistency of predicted and observed event frequencies | Rare high-probability bins need uncertainty intervals and sample counts. |
| Reliability and resolution diagnostics | Calibration versus discrimination | Better calibration alone does not establish more meteorological information. |
| Precision-recall analysis for rare events | Event screening at practical false-positive rates | Report observed event prevalence and selected operating points. |
| Event-level loss / net benefit | Whether calibration leads to better decisions for explicit costs | Loss parameters and mitigation efficacy must be documented and varied. |

| Metric | Why it matters | Scientific guardrail |
|---|---|---|
| Stratification by lead, season, region and hazard | Generalisation and failure analysis | Do not obscure weak sites or seasons with one pooled metric. |

Block bootstrap uncertainty intervals by weather episode, forecast issue or another documented dependence unit. Do not randomly split nearby forecast cycles from the same storm across training and test. Report the number of observed extreme events, not only the number of forecast cases. Neither a paceval speedup nor a reduction in Brier loss is assumed ahead of testing [R9, R14].

### 6.3 Numerical performance and fair attribution

Measure: (a) graph creation time, (b) warm single-case latency, (c) batch throughput, (d) p50/p95/p99 execution time, (e) peak memory, (f) CPU use and, where hardware measurement is available, (g) energy per million formula evaluations. Repeat runs under fixed clock, core affinity and reproducible compiler/runtime configurations. Present confidence intervals rather than one best-case timing.

Report both **mathematical-kernel speed** and **total-pipeline speed**. Compute the creation-amortisation threshold where relevant:

$$N_{\text{break-even}} = \frac{T_{\text{create}}}{T_{\text{native/case}} - T_{\text{paceval/case}}}$$

This expression applies only when the denominator is positive and the measurement model is comparable. If paceval is slower per case, creation overhead cannot be amortised into a speed advantage. Parallelization, SIMD, memory layout, formula complexity and batch size must be disclosed.

**Meaningful negative result.** A fair benchmark may show that optimized native C++ or another established postprocessing library is faster. This would not invalidate the paceval research question about formula reuse and auditability, but it would limit claims of high-performance superiority for this meteorological workload.

## 7. Validation, error policy and reproducibility

### 7.1 Independent mathematics reference

Write separate Python and native C++ reference implementations for the smoothed exceedance logit, calibrated probability, Brier term, derivatives and decision margin. Use high-precision evaluation for selected stress inputs and cross-check analytical derivatives against finite differences away from non-differentiable policy boundaries. Verify result ordering and input-array stride in the paceval batch API. Keep a golden file with input vectors, coefficients, expected outputs, units and model-version hashes.

**Proposed numerical acceptance:** On ordinary finite double-precision inputs, the relative or absolute difference in each bounded probability and decision expression should be below a pre-registered tolerance appropriate to conditioning, initially `abs(error) <= 1e-11 * max(1, abs(reference))`. Tighten or loosen only with a documented numerical analysis. For stress tests near underflow, overflow and small denominators, use case-specific policies and always report failures. This is an intended criterion, not a completed verification.

### 7.2 Forecast reproducibility controls

- Freeze model cycles, archive snapshots, site list, event thresholds and observation QC policies before evaluating held-out data.

- Ensure fitted coefficients and standardisation constants originate only from the relevant training period and training sites.
- Clearly distinguish statistical uncertainty of forecast probabilities from arithmetic rounding intervals.
- Include both no-action and action baseline policies for economic comparisons. Do not use made-up event losses in a claimed empirical result.
- Report source licences, attribution and whether numerical examples or raw station observations can legally be redistributed.
- Avoid treating forecasts from one issue or neighbouring ensemble members as independent statistical observations for confidence intervals.

## 8. Research gates, decision criteria and deployment maturity

| Gate | Positive demonstration | Stop or redesign if… |
|---|---|---|
| G1. Data contract | Archived forecasts and QC observations align at verified valid times and windows. | Accumulation windows, sensor definitions or timestamps cannot be matched credibly. |
| G2. Mathematics | Independent double and selected high-precision references match paceval within documented tolerances. | Results or derivatives disagree materially, or SDK edge cases are unhandled. |
| G3. Forecast skill | Held-out event calibration is competitive with, or better than, an appropriate raw and climatological baseline. | Apparent skill disappears on independent seasons, locations or rare-event strata. |
| G4. Compute fitness | Formula reuse, throughput, precision or auditability justifies its observed system cost. | Graph or data-movement overhead dominates, or optimized C++ performs better without offsetting engineering value. |
| G5. Industrial decision | Explicit, plausible cost and effectiveness assumptions yield useful and robust action margins. | Small coefficient changes flip recommendations or policies require unavailable physical-impact models. |
| G6. Operational governance | Provenance, versioning, human review and failure handling are documented. | The demonstration is confused with an authorised warning service or a certified safety function. |

The **minimum publishable research result** is a complete, reproducible comparison showing the correct numerical use of paceval, weather forecast verification against appropriate baselines, and accurate attribution of timing. A defensible claim of industrial superiority requires evidence in addition to completion of the demonstration.

## 9. Implementation roadmap

**Phase I. Mathematics and synthetic reproducibility.** Implement reusable graphs for calibration, Brier contributions, action margin, cost sensitivity and selected compound-event bounds. Add independent references and graph-lifecycle tests.

**Phase II. One historical forecast region.** Select a NOAA GEFS archive with suitable station

observations or another legitimately accessible hindcast archive. Freeze a few lead-time and valid-window definitions and complete proper out-of-time validation.

**Phase III. Germany operational replay.** Adapt ingestion to DWD ICON ensemble data and DWD station measurements. Use live processing as an engineering integration test until enough historical observations exist for external forecast-skill claims.

**Phase IV. Industrial pilot.** Agree on actual asset tolerances, approval workflow and economically credible action costs with an infrastructure partner. Measure end-to-end latency and false-alarm consequences under supervised evaluation. No automatic safety-critical actuation is authorised by this research plan.

## 10. Conclusions

This meteorological use case is a strong **research demonstration of paceval's fundamental role as a reusable mathematical engine**. It naturally involves high-volume, frequently repeated probability calibrations, nonlinear expressions, explicit cost functions, sensitivity calculations, versioned formula policies and independent numerical verification. The availability of public weather-model documentation and observations makes experimental design more transparent than in many proprietary industrial domains.

The scientifically credible positioning is precise: **paceval evaluates the mathematical post-processing and decision layer; it does not run the atmospheric forecasting model, nor does its numerical speed establish forecast skill.** Any performance or decision-quality improvement must be measured against competitive methods, with matched forecast events and well-defined costs.

## References and source provenance

The following publications and technical resources were checked for this research specification on **14 November 2026**. They provide background and data documentation. They are not external endorsements of paceval performance or this unimplemented proposal.

[R1] ECMWF. Open data: IFS and AIFS forecasts. Includes the October 2026 change to IFS 0.1-degree publication and rolling archive limitations.

[R2] ECMWF Forecast User Guide. Medium-range ensemble forecasts. Ensemble structure and forecast timing.

[R3] ECMWF. ECMWF open data: technical specifications. GRIB2 formats, streams, model cycles and resolutions.

[R4] NOAA NCEI. Global Ensemble Forecast System (GEFS). Forecast members and data-access pathways.

[R5] Deutscher Wetterdienst. Open numerical weather prediction directories. ICON, ICON-EPS and regional ensemble entries.

[R6] Deutscher Wetterdienst Climate Data Center. CDC Open Data introduction. Station observations, versioning, quality-control stages and metadata.

[R7] NOAA NCEI. Global Hourly station data. Historical observations and station access.

[R8] Copernicus Climate Change Service. Climate reanalysis and ERA5. Reanalysis characteristics and temporal coverage.

[R9] WMO. Forecast verification: reliability diagrams, Brier score and skill. Probability verification terminology.

[R10] NOAA National Weather Service. Impact-Based Decision Support Services. Operational decision-support framing.

[R11] Gneiting, T., Raftery, A. E., Westveld, A. H. III, and Goldman, T. (2005). Calibrated Probabilistic Forecasting Using Ensemble Model Output Statistics and Minimum CRPS Estimation.

*Monthly Weather Review*, 133, 1098–1118.

[R12] ECMWF. Calibration of Medium-Range Weather Forecasts. Statistical postprocessing and dependence considerations.

[R13] Finnish Meteorological Institute study. Operational statistical postprocessing of temperature ensemble forecasts with station-specific predictors. Operational station-specific calibration.

[R14] ECMWF Forecast User Guide. Verification metrics. Brier, reliability and probabilistic verification guidance.

[R15] ECMWF Forecast User Guide. Consideration of probability. Dependence across time windows and compound events; probability calibration.

[R16] ECMWF Forecast User Guide. Point rainfall and ecPoint. Site-scale rainfall variability and postprocessing context.

[R17] paceval. Official Mathematical Engine API documentation. `paceval_CreateComputation`, `paceval_dGetComputationResult`, `paceval_dGetComputationResultExt`.

[R18] paceval. Trusted Interval Computation (TINC). Numerical-computation interval feature.

[R19] ECMWF. ECMWF open data parameter reference. Surface variables `tp`, `10fg`, `2t` and available event probabilities.

[R20] WMO. Global Status of Multi-Hazard Early Warning Systems 2025. Broader social context, not product validation.

[R21] ECMWF. Forecast verification using information and noise (Technical Memorandum 927, 2025). Distinguishes calibration and resolution.

[R22] NWS. Probabilistic Decision Support. Example operational communication of percentile and exceedance guidance.

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**Repository status:** Research proposal. No C++ demonstrator, historical verification dataset, performance measurements or operational forecast product is included with this whitepaper. The proposed equations, experiments and success criteria are intended for subsequent implementation and independent verification.